Hawking's proposal that black holes radiate didn't come out of the blue. It came in response to a suggestion from Jacob Bekenstein—who at the time was yet another graduate student of John Wheeler's at Princeton—that black holes should have entropy.

One of the motivations behind Bekenstein's idea was the fact that, according to classical general relativity, the area of a black hole's event horizon can never decrease. That sounds suspiciously like the second law of thermodynamics, according to which the entropy of a closed system can never decrease. Inspired by this similarity, physicists constructed an elaborate analogy between the laws of thermodynamics and the behavior of black holes, according to which the mass of the black hole is like the energy of a thermodynamic system, and the area of the event horizon is like the entropy.

Bekenstein suggested that it was more than an analogy. The area of the event horizon isn't just *like* the entropy, it *is* the entropy of the black hole, or at least proportional to it. Hawking and others scoffed at the suggestion at first—if black holes have entropy like conventional thermodynamic systems, they should also have a temperature, and then they should give off radiation! Motivated to disprove this ridiculous-sounding notion, Hawking ended up showing that it was all true. These days we refer to the entropy of a black hole as the *Bekenstein-Hawking entropy*.

One reason why this is such a provocative result is that classically, black holes don't seem like things that should have entropy at all. They're just regions of empty space. You get entropy when your system is made of atoms or other tiny constituents, which can be arranged in many different ways while maintaining the same macroscopic appearance. What are these constituents supposed to be for a black hole? The answer has to come from quantum mechanics.

It's natural to presume that the Bekenstein-Hawking entropy of a black hole is a kind of entanglement entropy. There are some degrees of freedom inside the black hole, and they are entangled with the outside world. What are they?

We might first guess that the degrees of freedom are simply vibrational modes of the quantum fields inside the black hole. There are a couple of problems with that. For one thing, the real answer for the entropy of a region in quantum field theory was "infinity." We could wrestle that down to a finite number by choosing to ignore very-small-wavelength modes, but